Robust FLOWER Under Noise Model Mismatch

A controlled study of reconstruction quality

Itay Shorian

Generative Learning and Diffusion Models

October 2026

FLOWER: From a Measurement to a Clean Image

Inverse Problem

We observe a degraded, noisy image and seek the underlying clean image.

The Image Prior

A pretrained flow model predicts a clean destination from the current trajectory state.

$$y = Hx + n, \qquad n \sim \mathcal{N}(0, \sigma_{\text{true}}^2 I)$$

Three Steps at Every Flow Time

1. Estimate the clean destination
2. Refine it using the measurement
3. Move to the next flow time

The Noise Level Sets the Refinement Strength

Destination Estimate

$$X_t = (1-t)X_0 + tX_1, \qquad \hat{x}_1(x_t) = x_t + (1-t)v_\theta(x_t, t)$$

Measurement-Aware Refinement

$$\mu_t = \arg\min_x \left\{ \frac{\|Hx-y\|_2^2}{2\sigma_a^2} + \frac{\|x-\hat{x}_1\|_2^2}{2\nu_t^2} \right\}$$

Interpretation

A smaller assumed σ gives more weight to the measurement.
A larger assumed σ gives more weight to the predicted destination.

The paper's image benchmark uses Flower1-OT with γ = 0

Our Research Question

Noise Calibration

How does reconstruction quality change when FLOWER receives an incorrect noise standard deviation?

$$r = \sigma_{assumed} / \sigma_{true}$$

r	0.50	0.75	1.00	1.25	1.50
Variance ratio	0.25	0.5625	1.00	1.5625	2.25

Paired Comparison

The image, measurement and source-noise draws are shared across all five ratios. We compare every result with its r = 1 control.

Experimental Design

20 fixed CelebA images + 20 fixed AFHQ-Cat images

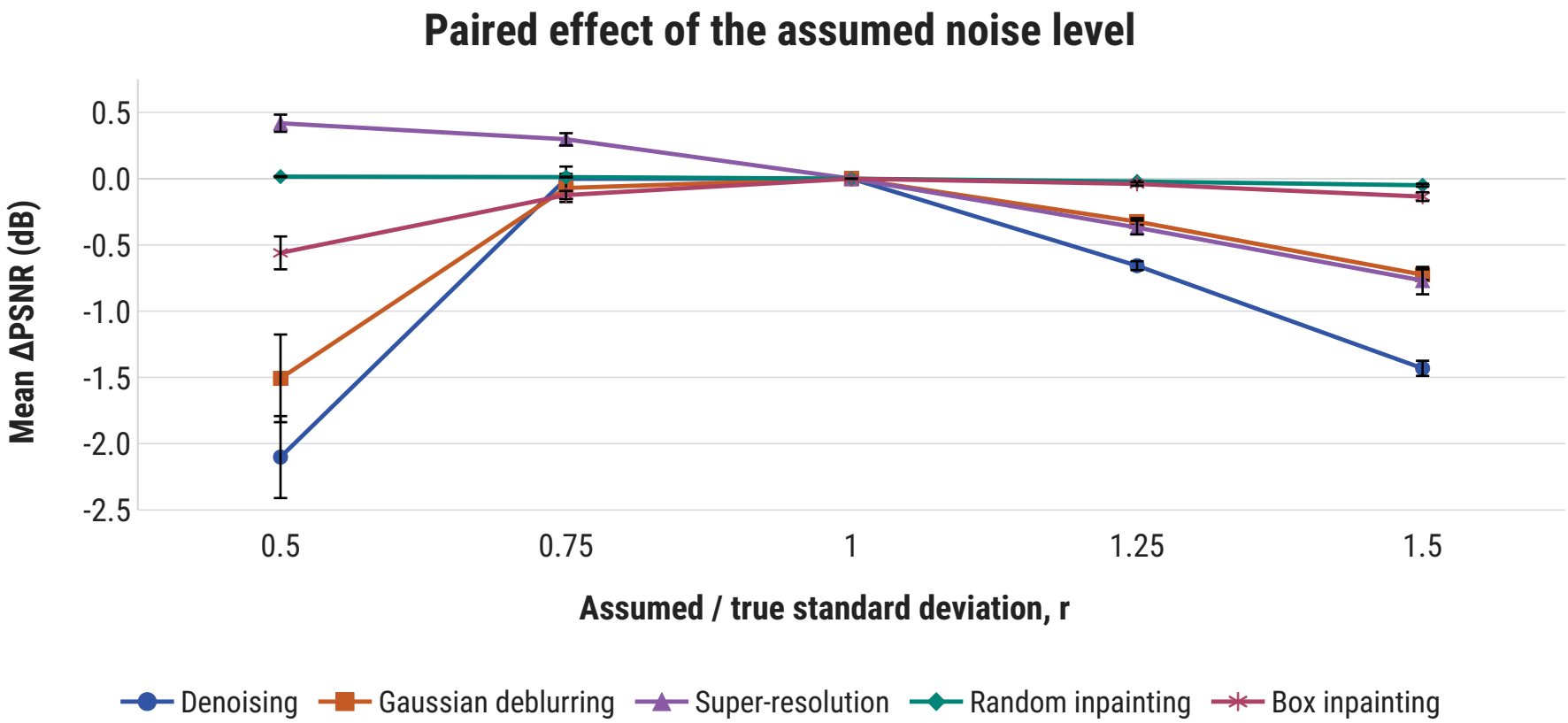
Inverse problem	True σ	CelebA	AFHQ-Cat
Denoising	0.20	Identity	Identity
Gaussian deblurring	0.05	61 × 61, blur σ = 1	61 × 61, blur σ = 3
Super-resolution	0.05	2× subsampling	4× subsampling
Random inpainting	0.01	70% missing	70% missing
Box inpainting	0.05	40 × 40 central box	80 × 80 central box

40 images × 5 problems × 5 ratios = 1,000 reconstructions

PSNR ↑ SSIM ↑ LPIPS ↓

Same selected images throughout each dataset. Seed 2026. Noise σ is in model coordinates [-1, 1].

PSNR Response: CelebA

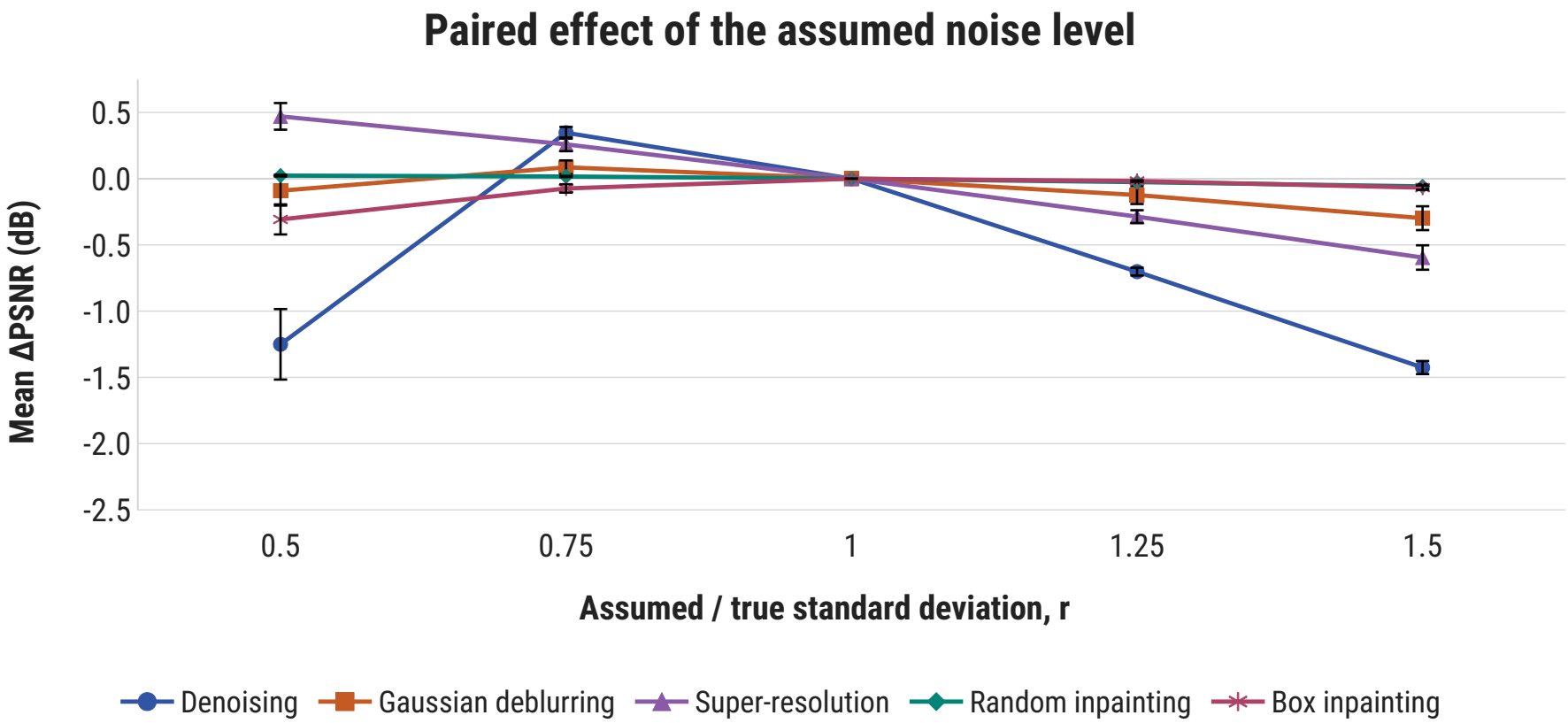


Main Observation

Denoising loses 2.10 dB at r = 0.50. Super-resolution gains 0.42 dB. Random inpainting changes very little.

Mean paired ΔPSNR vs r = 1. Higher is better. n = 20. Error bars: approximate 95% CI.

PSNR Response: AFHQ-Cat

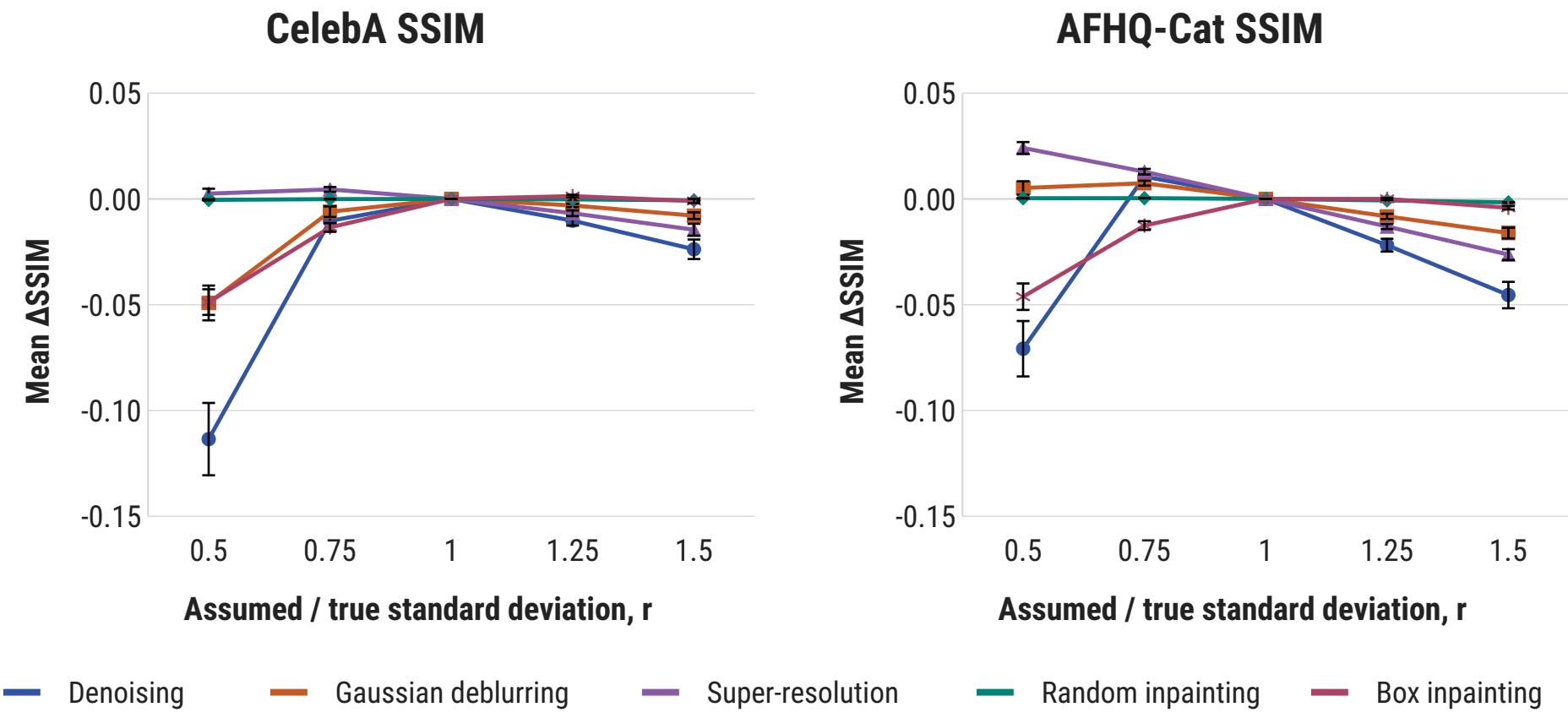


Main Observation

Denoising improves by 0.35 dB at r = 0.75, but loses 1.25 dB at r = 0.50. Super-resolution gains 0.47 dB at r = 0.50.

Mean paired ΔPSNR vs r = 1. Higher is better. n = 20. Error bars: approximate 95% CI.

SSIM Response Across Both Datasets

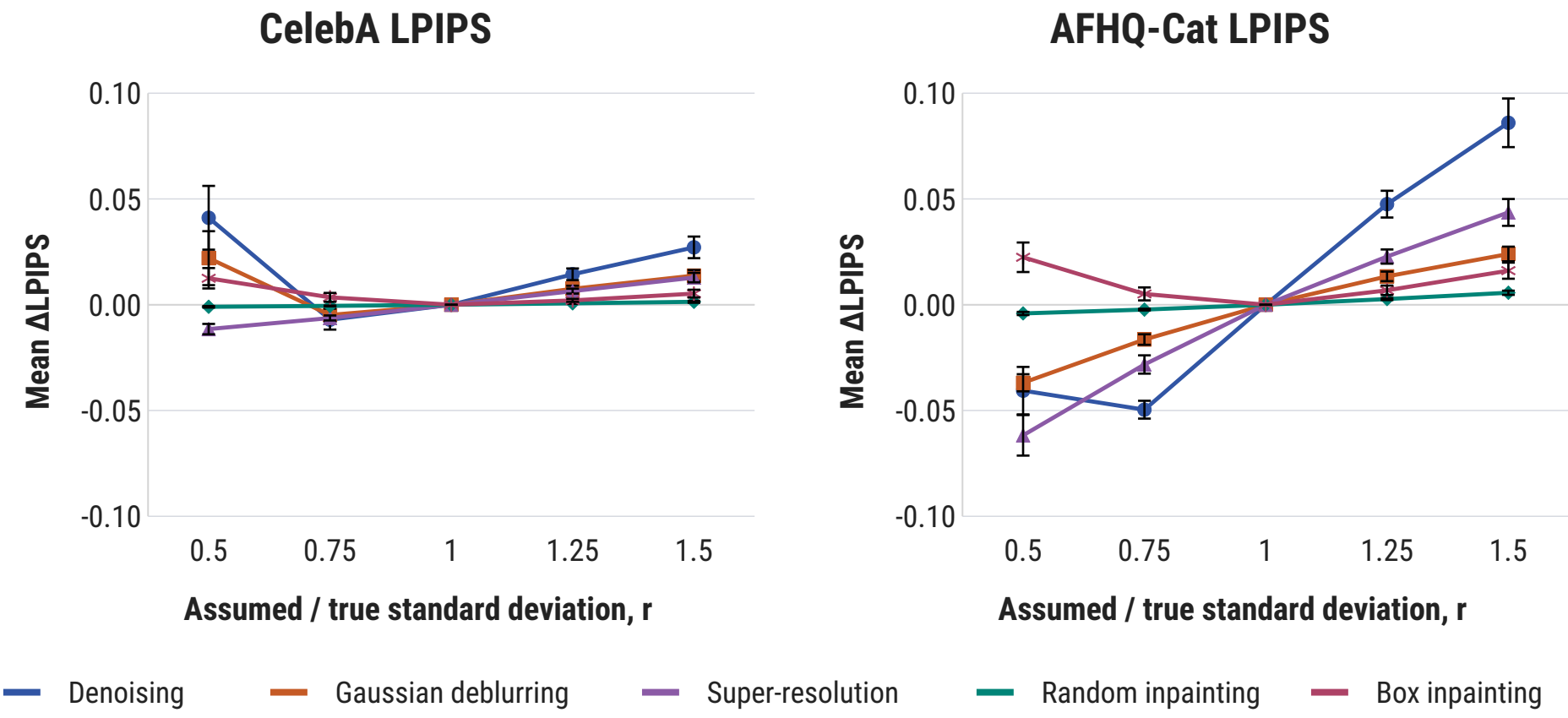


What the Metric Adds

Underestimating noise can damage structure even when PSNR changes little. Super-resolution improves in both datasets.

Mean paired change vs r = 1. Higher is better. n = 20. Error bars: approximate 95% CI.

LPIPS Response Across Both Datasets



What the Metric Adds

At r = 0.50, AFHQ-Cat denoising improves LPIPS while PSNR and SSIM worsen. Perceptual and pixel fidelity can disagree.

Mean paired change vs r = 1. Lower is better. n = 20. Error bars: approximate 95% CI.

All Tasks: Mean PSNR Change (dB)

Cells compare a mismatched reconstruction with its r = 1 control

CelebA

Task	0.50	0.75	1.25	1.50
Denoising	-2.10	0.00	-0.66	-1.43
deblurring	-1.51	-0.07	-0.32	-0.72
Super-resolution	+0.42	+0.30	-0.37	-0.77
Random inp.	+0.02	+0.01	-0.02	-0.05
Box inp.	-0.56	-0.12	-0.04	-0.14

AFHQ-Cat

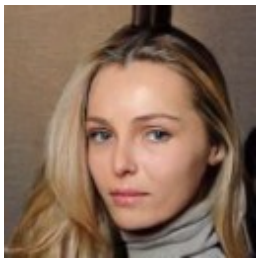
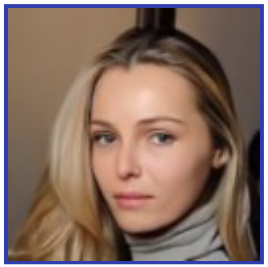
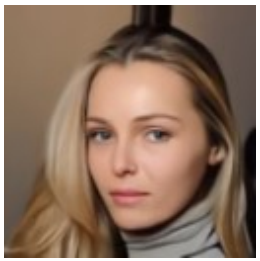
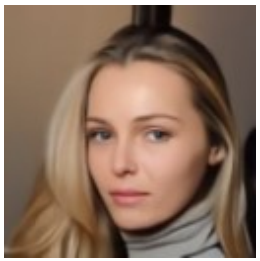
Task	0.50	0.75	1.25	1.50
Denoising	-1.25	+0.35	-0.70	-1.43
deblurring	-0.09	+0.09	-0.12	-0.30
Super-resolution	+0.47	+0.26	-0.29	-0.60
Random inp.	+0.02	+0.02	-0.03	-0.06
Box inp.	-0.31	-0.07	-0.02	-0.07

Reading the Table

Positive values mean better PSNR. Overestimation lowers the mean in every tested dataset–task setting.

Each cell averages 20 paired image differences. The r = 1 column is zero by definition.

Visual Comparison: Denoising

Ground truth	Measurement	r = 0.50	r = 0.75	**r = 1.00**	r = 1.25	r = 1.50
CelebA						
						
		29.72 / 0.803 0.097	31.77 / 0.913 0.033	31.85 / 0.928 0.020	31.34 / 0.920 0.028	30.70 / 0.909 0.037
AFHQ-Cat						
						
		29.78 / 0.813 0.043	30.87 / 0.868 0.057	30.57 / 0.846 0.111	29.89 / 0.812 0.162	29.15 / 0.774 0.203

Observation

Underestimation preserves stronger fine-scale variation. Overestimation smooths it. The metrics determine whether this is detail or residual noise.

Numbers: PSNR (dB) / SSIM, then LPIPS. Fixed examples; blue frame = correct noise. Measurement = $H^T y$.

Visual Comparison: Gaussian deblurring

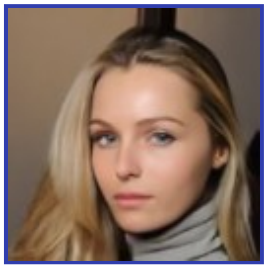
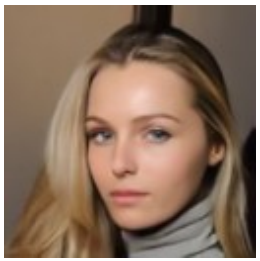
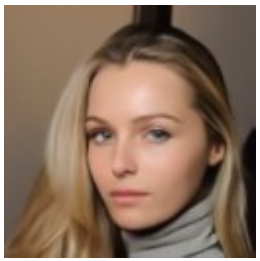
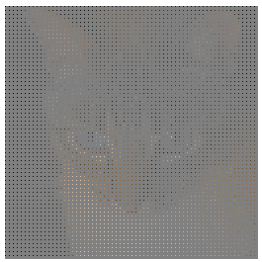
Ground truth	Measurement	r = 0.50	r = 0.75	**r = 1.00**	r = 1.25	r = 1.50
CelebA						
						
		31.64 / 0.866 0.075	32.94 / 0.925 0.026	33.12 / 0.941 0.019	32.86 / 0.939 0.025	32.53 / 0.934 0.030
AFHQ-Cat						
						
		26.17 / 0.724 0.296	26.45 / 0.722 0.310	26.52 / 0.712 0.326	26.34 / 0.699 0.339	26.10 / 0.685 0.346

Observation

The two datasets respond differently to underestimation. Kernel width and image content differ between them.

Numbers: PSNR (dB) / SSIM, then LPIPS. Fixed examples; blue frame = correct noise. Measurement = $H^T y$.

Visual Comparison: Super-resolution

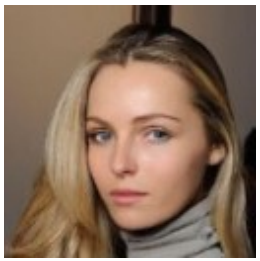
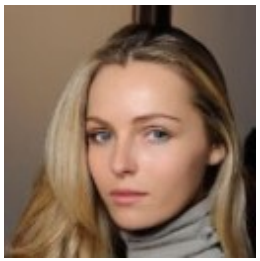
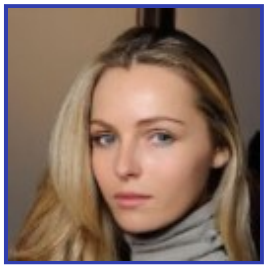
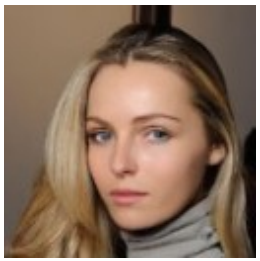
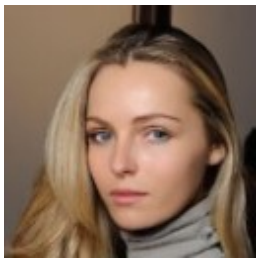
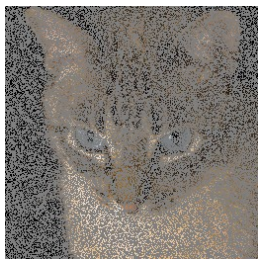
Ground truth	Measurement	r = 0.50	r = 0.75	**r = 1.00**	r = 1.25	r = 1.50
CelebA						
						
		31.34 / 0.921 0.016	31.29 / 0.924 0.018	31.13 / 0.922 0.022	30.90 / 0.917 0.026	30.64 / 0.911 0.030
AFHQ-Cat						
						
		25.57 / 0.663 0.247	25.46 / 0.643 0.271	25.33 / 0.624 0.292	25.16 / 0.606 0.308	24.97 / 0.590 0.324

Observation

Both datasets gain PSNR and SSIM and reduce LPIPS at r = 0.50 in the 20-image average.

Numbers: PSNR (dB) / SSIM, then LPIPS. Fixed examples; blue frame = correct noise. Measurement = $H^T y$.

Visual Comparison: Random inpainting

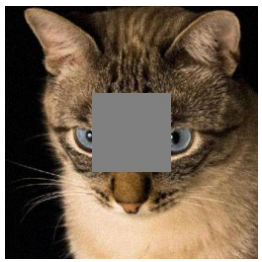
Ground truth	Measurement	r = 0.50	r = 0.75	**r = 1.00**	r = 1.25	r = 1.50
CelebA						
						
		32.06 / 0.939 0.014	32.06 / 0.939 0.014	32.05 / 0.940 0.015	32.04 / 0.940 0.015	32.03 / 0.939 0.015
AFHQ-Cat						
						
		30.58 / 0.892 0.049	30.57 / 0.892 0.051	30.57 / 0.891 0.052	30.56 / 0.890 0.054	30.54 / 0.888 0.057

Observation

The reconstruction changes only slightly over this range. This task also has the smallest true noise level, σ = 0.01.

Numbers: PSNR (dB) / SSIM, then LPIPS. Fixed examples; blue frame = correct noise. Measurement = $H^T y$.

Visual Comparison: Box inpainting

Ground truth	Measurement	r = 0.50	r = 0.75	**r = 1.00**	r = 1.25	r = 1.50
CelebA						
						
		31.40 / 0.905 0.044	31.97 / 0.939 0.029	32.18 / 0.958 0.012	32.10 / 0.959 0.014	31.95 / 0.958 0.016
AFHQ-Cat						
						
		23.82 / 0.855 0.081	23.89 / 0.886 0.077	23.90 / 0.895 0.079	23.88 / 0.893 0.085	23.86 / 0.887 0.094

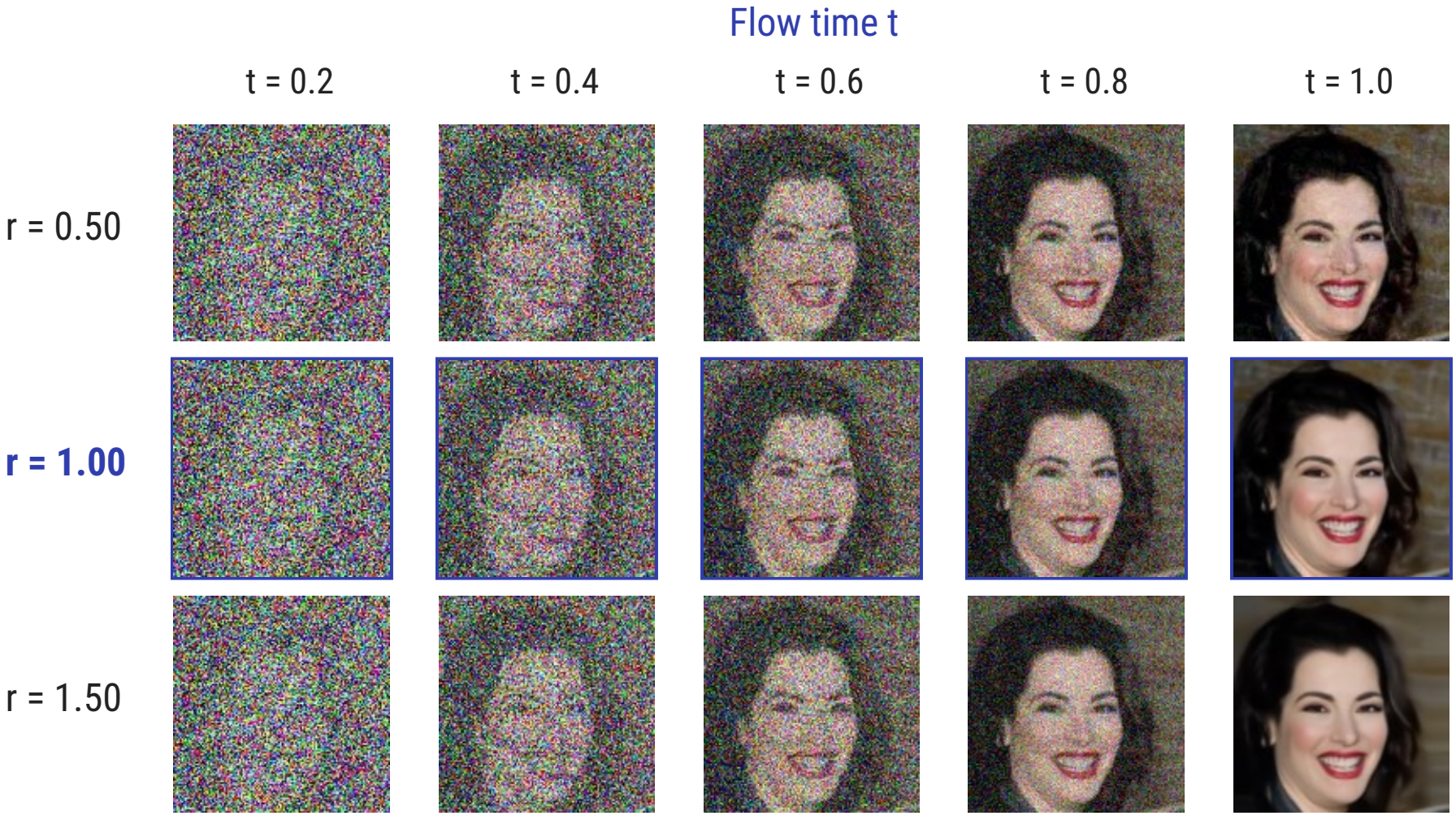
Observation

The data term acts on observed pixels. The prior supplies the missing central region, and later iterations couple both regions.

Numbers: PSNR (dB) / SSIM, then LPIPS. Fixed examples; blue frame = correct noise. Measurement = $H^T y$.

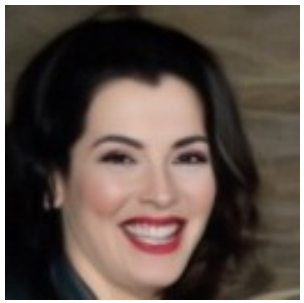
How Mismatch Changes the Trajectory

CelebA denoising, image 182734.jpg



Rows share measurement and source-noise draws. Final column: reconstruction. All five ratios are in the report.

Inside the Three Steps at t = 0.75

	Destination estimate	Measurement refinement	Next trajectory state
r = 0.50			
r = 1.00			
r = 1.50			

CelebA denoising, image 182734.jpg. Step 2 uses σ directly. Step 3 adds source noise and advances time.

Interpreting the Evidence

Why the Forward Operator Matters

Denoising changes the measurement weight at every pixel. Masked and subsampled problems constrain only observed locations directly.

Why Correct Noise Need Not Maximize PSNR

The practical solver uses a learned velocity and an approximate endpoint distribution. A different data weight can compensate for their errors on a finite sample.

Limits of the Comparison

20 images per dataset and one paired seed sequence per image–task pair. Paper results use 100 images. Confidence intervals summarize image variation.

These results establish sensitivity on the tested sample; they do not identify a universal optimal noise ratio.

Conclusions

Sensitivity Depends on the Task

Denoising is the most sensitive setting. Random inpainting is nearly unchanged over the tested range.

Overestimation Consistently Hurts Mean PSNR

Both r = 1.25 and r = 1.50 reduce mean PSNR in all ten dataset–task settings.

Underestimation Has No Universal Outcome

Super-resolution improves in both datasets. Severe underestimation hurts denoising, and LPIPS can disagree with PSNR and SSIM.

Noise calibration should be evaluated for the task and the metric.

References and Reproducible Results

M. Pourya, B. El Rawas and M. Unser
FLOWER: A Flow-Matching Solver for Inverse Problems
ICLR 2026

Y. Lipman et al.
Flow Matching for Generative Modeling
ICLR 2023

Project Results

Raw and paired metrics, all 50 summary rows, saved examples, and trajectory captures are in artifacts/experiments/main/.

github.com/ItayShorian07/Diffusion--flower-project